Chawin Mingsuwan

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Education

Purdue University West Lafayette, IN
Bachelor of Science in Video Game Development *May 2027*
Minors in Advanced Global Technology, Earth and Planetary Science; Certification in Data Science GPA: 3.96
Dean's List Fall 2023 - Fall 2025

Professional Experience

Johns Hopkins Applied Physics Laboratory (APL) — NASA JSC XR Mission Control Environment West Lafayette, IN
***Technical Lead, XR & AI Systems* | Sponsored Project** *October 2025 - Present*

- Led development of a **Unity-based immersive Virtual Agent** using a **Retrieval-Augmented Generation (RAG) pipeline** to support future NASA Mission Control workflows and decision-making.
- **Independently identified, contacted, and interviewed NASA Flight Directors, FDOs, and astronauts** to gather real operational knowledge, translating qualitative insights into training data and system requirements for AI personas.

NASA/Barrios - Purdue Datamine Corporate Partner West Lafayette, IN
Lead User-Experience (UX) and Virtual Reality (VR) Unreal Engine Developer *January 2025 - December 2025*

- Led the integration of all interactive systems, **rotation controls, dynamic color-coding, multi-camera logic, and UI panel interactions** by unifying **Blueprint** and **C++** behavior for mission planning and astronaut training.
- Built a **Meta Quest 2 VR simulation** of the **NASA Gateway station in Unreal Engine 5**, enabling **early astronaut training use-cases**, including EVA planning and spatial orientation.

Los Alamos National Laboratory Los Alamos, NM
Extended Reality (XR) and High-Performance Computing (HPC) Intern *May 2025 - August 2025*

- **Engineered immersive XR environments** in Unreal Engine 5 to **visualize high-performance computing simulations** from **ParaView** to enable scientific research.
- Implemented first-of-its-kind **Apple Vision Pro hand gesture interactions** using **C++ and Blueprints**, applying **3D joint vector math (distances and thresholds)** to detect gestures and trigger real-time UI interactions.

Ilender Peru Internship Lima, Peru
Software Engineering Intern *May 2024 - July 2024*

- **Automated multi-country data** workflows (Peru, Bolivia, Mexico, Colombia, Venezuela) in **Python** and built an **ARIMA forecasting model**, improving data processing efficiency by **30%** and prediction accuracy by **20%**.

Projects and Publications

NASA/IEEE Publication — Foundations of a Visualization Tool for NASA Gateway and Lunar Surface Operations

- **First Author**, peer-reviewed NASA/IEEE publication presented at the **2025 IEEE SMC-IT Conference**, demonstrating **XR visualization for Gateway mission planning**.

Tidal Disruption Event (TDE) Astrophysics Pipeline

- Cleaned and processed ALeRCE/ZTF light-curve data in **Python** to identify TDE candidate signatures, including the characteristic **$t^{-5/3}$ decay pattern**.
- Built and deployed a full-stack visualization pipeline in **Python**, implementing a **FastAPI backend** and frontend with **JavaScript/React** that generates **Matplotlib light-curve plots**.

Virtual Reality Study Room Prototype

- **Led development** of an interactive VR Study Room prototype in **Unity**, programming core **C# mechanics** to enhance **user engagement** and improve **accessibility** for alternative learning methods.
- Implemented **Unity XR collaborative features**, including virtual whiteboard writing to simulate real-time tutoring and teaching in VR, resulting in the **most highly rated prototype** for interactivity and instructional realism.

Leadership Experience

Purdue Space Day West Lafayette, IN
Director of Activities and Treasurer *October 2024 - Present*

- Coordinated **11+ orgs**, **400+ volunteers**, and **9 STEM activities** for **600 students**; first to integrate the **NASA Gateway Visualization Tool (GVT)** into the event.